Web Scraping Exercises

Part I: Multiple Choice

1. What does CSV stand for?

A. Computer Storage Values
B. Comma-Separated Values
C. Column Structured Variables
D. Common SQL Values

2. In a CSV file, each line typically represents:

A. A database
B. A column
C. A record
D. A primary key

3. Which SQL statement is used to import data from a CSV file into a MySQL table?

A. IMPORT DATA
B. INSERT FILE
C. LOAD DATA
D. READ CSV

4. Which SQL statement can be used to export data from a MySQL table into a CSV file?

A. EXPORT TABLE
B. SELECT ... INTO OUTFILE
C. SAVE TABLE
D. LOAD DATA

5. What does the following clause indicate?

```
FIELDS TERMINATED BY ','
```

A. Each record ends with a comma.
B. Each field is separated by a comma.
C. Each column must contain a comma.
D. Each field must begin with a comma.

6. What is the purpose of the following clause when importing a CSV file?

```
IGNORE 1 ROWS
```

A. Ignore records containing NULL values.
B. Ignore duplicate records.
C. Skip the first row of the CSV file.
D. Skip records containing errors.

7. What does the following clause specify?

```
OPTIONALLY ENCLOSED BY '"'
```

A. Every value must contain quotation marks.
B. Fields may be enclosed in double quotation marks.
C. Double quotation marks are removed from the database.
D. Only numeric values are imported.

8. What is web scraping?

A. A method of deleting unwanted webpages.
B. A technique for designing websites.
C. The process of automatically extracting information from websites.
D. A method of protecting websites from hackers.

9. Beautiful Soup is primarily used with which programming language?

A. Java
B. Python
C. C++
D. SQL

10. Beautiful Soup is commonly used to extract information from:

A. HTML and XML documents
B. Executable programs
C. MySQL database files
D. PDF documents only

Part II: True or False

11. CSV files can be opened and edited using a text editor.

12. A CSV file must always contain column headings in its first row.

13. An empty field in a CSV file is always imported into MySQL as NULL.

14. MySQL Workbench provides a graphical interface for importing CSV files.

15. When using `LOAD DATA LOCAL INFILE`, local file loading may need to be enabled on both the MySQL server and client.

16. SQL NULL and an empty string (`''`) represent the same thing.

17. Web scraping eliminates the need to manually copy information from websites.

18. Information collected through web scraping can be stored in a MySQL database.

19. Web scraping can be used to extract information from any website without restrictions.

20. All string values in a CSV file must be enclosed in double quotation marks.